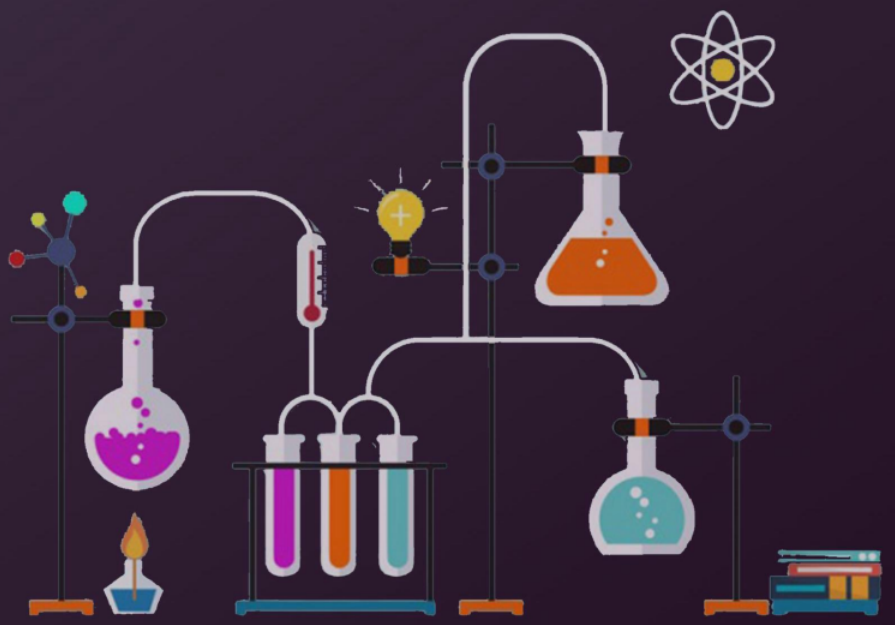
CHEMISTRY

s-Block



SHEET

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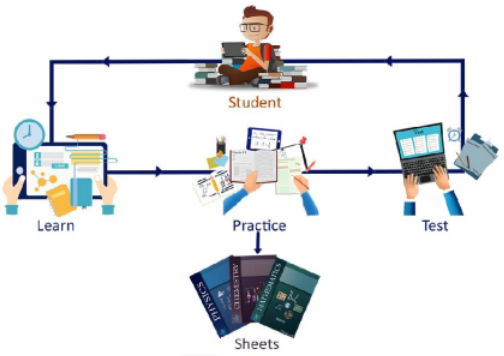
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to s-block elements	1-2			
2	Physical And Chemical Properties Of Alkali Metals	3-16	1,2,6-8, 13,17	1-2	1-3,19
3	Alkali Metal oxides and halides, anomalous behaviour of Lithium	17-24	3,14	3-8	4,14, 16-18,20
4	Important Compounds Of Sodium And Potassium	25-30	12	9-15	
5	Physical And Chemical Properties Of Alkali Earth Metals	31-33	4	16-21	8-12
6	General Characterstics of Alkali Earth Metal Compounds	34-42	5,9-11	22	5
7	Important Compounds Of Calcium	43-45	15	23-25	6,7
8	ACC	46-50	16,18	26-31	13,15

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to s-Block Elements

Q1. Total number of s-block elements is are –
(A) 14 (B) 13
(C) 12 (D) 11

Q2. The radioactive s-block elements are–
(A) Ra and Ba (B) Ra and Fr
(C) Fr and Cs (D) Rb and Sr

Physical and Chemical Properties of Alkali Metals

Q3. Which of the following has the maximum solubility in water?
(A) $LiHCO_3$ (B) $NaHCO_3$
(C) $KHCO_3$ (D) $RbHCO_3$

Q4. The s-block elements of the periodic table are those in which the last electron enters the s-subshell of the _____ shell.
(A) penultimate
(B) antepenultimate
(C) outermost
(D) None

Q5. Which of the following alkali metals has the lowest melting point?
(A) Na (B) Li
(C) K (D) Cs

Q6. Which of the following has the lowest density ?
(A) Be (B) Mg
(C) Ca (D) Sr

Q7. Which of the following is incorrect ?
(A) Alkaline earth metals like alkali metals have high electrical and thermal conductivities.
(B) Alkaline earth metals are more electropositive (or metallic) than alkali metals.
(C) The electropositive (metallic) character of alkaline earth metals increases down the group.
(D) None of these

Q8. Which of the following is the most powerful reducing agent in the gas phase ?
(A) Cs (B) Li
(C) K (D) Na

Q9. Which of the following is soluble in organic solvents like ethanol ?
(A) LiCl (B) NaCl
(C) KCl (D) RbCl

Q10. The correct order of stability for the following superoxides is :-
(A) $KO_2 > RbO_2 > CsO_2$
(B) $RbO_2 > CsO_2 > KO_2$
(C) $CsO_2 > RbO_2 > KO_2$
(D) $KO_2 > CsO_2 > RbO_2$

Q11. Select the correct order of size?
(A) $Li > Na > K > Rb > Cs$
(B) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
(C) $Li^+_{(aq)} > Na^+_{(aq)} > K^+_{(aq)} > Rb^+_{(aq)} > Cs^+_{(aq)}$
(D) None of these

Q12. Among the following which one has the highest cation to anion size ratio :-
(A) CsF (B) LiF
(C) NaF (D) CsI

Q13. Which of the alkali metal forms only, the normal oxide M_2O ?
(A) Li (B) Na
(C) K (D) Rb

Q14. The pair of compounds, which cannot exist together in a solution is
(A) $NaHCO_3$ and NaOH
(B) Na_2CO_3 and NaOH
(C) $NaHCO_3$ and Na_2CO_3
(D) $NaHCO_3$ and H_2O

Q15. The order of solubility of lithium halides in non-polar solvents follows the order
(A) LiI > LiBr > LiCl > LiF
(B) LiF > LiI > LiBr > LiCl
(C) LiCl > LiF > LiI > LiBr
(D) LiBr > LiCl > LiF > LiI

Q16. Which of the following carbonate of alkali metals has the least thermal stability?
(A) Li_2CO_3 (B) K_2CO_3
(C) Cs_2CO_3 (D) Na_2CO_3

Alkali Metals Oxides and Halides, Anomalous Behaviour of Lithium

Q17. Chemically, group 1 elements are very reactive and tarnish rapidly in air due to the formation of :-
(A) oxides (B) hydroxides
(C) carbonates (D) all of these

Q18. Which of the following is incorrect for the hydroxides of alkali metals?
(A) They are white crystalline solids
(B) These caustic alkalies are the strongest bases known in aqueous solution.
(C) They readily dissolve in water with the evolution of much heat. The solubility in water decreases as we move down the group.
(D) The basic strength of hydroxides increases on moving down the group.

Q19. Alkali metal halides have high negative enthalpies of formation. The $\Delta_f H^\circ$ values for _____ become less negative as we go down the group.
(A) iodides (B) bromides
(C) chlorides (D) fluorides

Q20. Which of the following is incorrect about group 2 metals ?
(A) Except Be & Mg they dissolve in liquid NH_3.
(B) Dilute solutions of group 2 metal in liquid NH_3 are deep blue in color but the concentrated solutions are bronze colored.
(C) Solutions of group 2 Metals in liquid NH_3 decompose slowly on standing forming amides and evolving H_2.
(D) Dilute solution is diamagnetic in nature.

Q21. Potassium carbonate cannot be made by the Solvay process because
(A) potassium hydrogen carbonate is unstable
(B) potassium hydrogen carbonate is rather too soluble in water to be precipitated
(C) potassium carbonate is insoluble in water
(D) potassium carbonate is soluble in water

Q22. Quick lime is prepared on a commercial scale by heating _____ in a rotary kiln at 1070 – 1270 K.
(A) $CaSO_4$ (B) $Ca(NO_3)_2$
(C) $CaCO_3$ (D) $Ca(OH)_2$

Q23. Which of the following compound is efflorescent :-
(A) Washing soda (B) Caustic soda
(C) Caustic potas (D) Baking Soda

Q24. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then, A, B, C and D may be identified as :
(A) Na, H_2, NaOH and Zn
(B) K, H_2, KOH and Zn
(C) K, H_2, NaOH and Zn
(D) Ca, H_2, $CaCOH_2$ and Zn

Important Compounds of Sodium and Potassium

Q25. Which of the following is incorrect about lithium ?
(A) LiOH is a strong base.
(B) Unlike other alkali metals, Li does not react with ethyne to form lithium ethynide.
(C) Unlike other alkali metals, Li form dilithium acetylide on heating with C.
(D) Lithium has a grater tendency to form complexes than other alkali metals.

Q26. Baking soda is
(A) Na_2CO_3
(B) $NaHCO_3$
(C) Na_2SO_4
(D) K_2CO_3

Q27. The formula of Indian salt peter is
(A) KNO_3
(B) $NaNO_3$
(C) NaCl
(D) Na_2CO_3

Q28. Ripening of fruits can be carried out in presence of :-
(A) Na_2SO_4 (B) NaCl
(C) $CaCl_2$ (D) CaC_2

Q29. Fire extinguishers contain
(A) conc. H_2SO_4 solution
(B) H_2SO_4 and $NaHCO_3$ solutions
(C) $NaHCO_3$ solution
(D) $Ca1CO_3$ solution

Q30. CO_2 gas along with solid (Y) is obtained when sodium salt (X) is heated. (X) is again obtained when CO_2 gas is passed into :-
(A) Na_2CO_3, Na_2O
(B) Na_2CO_3, NaOH
(C) $NaHCO_3$, Na_2CO_3
(D) Na_2CO_3, $NaHCO_3$

Physical and Chemical Properties of Alkali Earth Metals

Q31. The incorrect statement is :-
(A) $BeCl_2$ is a covalent compound
(B) $BeCl_2$ can form dimer
(C) $BeCl_2$ is an e^--deficient molecule
(D) The hybrid state of Be in $BeCl_2$ is Sp^2

Q32. The 'milk of magnesia' used as an antacid is chemically
(A) $Mg(OH)_2$
(B) MgO
(C) $MgCl_2$
(D) MgO + $MgCl_2$

Q33. An alkaline earth metal (M) gives a salt with chlorine, which is soluble in water at room temperature. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone' -a white pigment. Metal M is -
(A) Ca (B) Mg
(C) Ba (D) Sr

General Characteristics of Alkali Earth Metal Compounds

Q34. Which of the following compound decomposes at highest temperature :-
(A) $SrCO_3$ (B) $BaCO_3$
(C) $CaCO_3$ (D) $MgCO_3$

Q35. Which of the following hydroxides of group 2 elements is amphoteric?
(A) $Mg(OH)_2$ (B) $Ca(OH)_2$
(C) $Be(OH)_2$ (D) $Sr(OH)_2$

Q36. Which of the following is incorrect for slaked lime?
(A) It is a white amorphous powder.
(B) It is sparingly soluble in water.
(C) The solubility increases with rise in temp.
(D) None of these

Q37. Plaster of Paris, a white powder, is -

(A) $CaSO_4.2H_2O$ (B) $CaSO_4.\frac{1}{2}H_2O$

(C) $CaSO_4$ (D) $CaSO_4.H_2O$

Q38. Gypsum is represented as

(A) $CaSO_4.H_2O$ (B) $CaSO_4$

(C) $2CaSO_4.2H_2O$ (D) $CaSO_4.2H_2O$

Q39. The wire of flash bulb is made of :-

(A) Mg (B) Cu

(C) Ba (D) Ag

Q40. The correct order of second ionisation potentials (IP) of Ca, Ba and K is

(A) K > Ca > Ba (B) Ba > Ca > K

(C) K > Ba > Ca (D) K = Ba = Ca

Q41. The correct order of basic-strength of oxides of alkaline earth metals is

(A) BeO > MgO > CaO > SrO

(B) SrO > CaO > MgO > BeO

(C) BeO > CaO > MgO > SrO

(D) SrO > MgO > CaO > BeO

Q42. The order of thermal stability of carbonates of IIA group is

(A) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$

(B) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

(C) $CaCO_3 > SrCO_3 > BaCO_3 > MgCO_3$

(D) $MgCO_3 = CaCO_3 > SrCO_3 = BaCO_3$

Important Compounds of Calcium

Q43. Which of the following carbonates is the most thermally stable ?

(A) $BeCO_3$ (B) $MgCO_3$

(C) $CaCO_3$ (D) $BaCO_3$

Q44. The main pigment for the absorption of light in green plants is chlorophyll-a which contains

(A) Be (B) Mg

(C) Ca (D) Sr

Q45. Ca^{2+} ions play an important role in

(i) neuromuscular function

(ii) interneuronal transmission

(iii) cell membrane integrity

(iv) blood coagulation

(A) (i), (ii) (B) (ii), (iii)

(C) (ii), (iii), (iv) (D) (i), (ii), (iii), (iv)

ACC

Q46. Which of the following is incorrect regarding the solutions of alkali metals in liquid NH_3?

(A) As the concentration increases above 3 M, the color of solution changes form dark blue to copper-bronze.

(B) Blue colored solutions as well as bronze-colored solutions are paramagnetic.

(C) These solutions conduct electricity better than any salt in any liquid and the conductivity is similar to that of pure metals.

(D) The solutions act as powerful reducing agents for the elements of groups 14, 15 and 16 for many compounds and coordination complexes. They even reduce an aromatic ring.

Q47. Unstable hardness of water due to presence of :

(A) $CaCl_2$, $MgSO_4$

(B) Ca^{+2}, Mg^{+2}

(C) $K^{\oplus}$, $CaCO_3$

(D) $Ca(HCO_3)_2$, $Mg(HCO_3)_2$

Q48. Total no. of moles of ions present in 1 molar aqueous solution of potash alum.

(A) 6 (B) 7

(C) 8 (D) 9

Q49. A metal which is soluble in both water and liquid NH_3 separately -

(A) Cr (B) Mn

(C) Ba (D) Al

Q50. Which of the following groups of elements have chemical properties that are most similar

(A) Na, K, Ca (B) Mg, Sr, Ba

(C) Be, Al, Ca (D) Be, Ra, Cs

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Sodium metal is highly reactive and can be stored under
(A) toluene
(B) kerosene oil
(C) alcohol
(D) benzene

Q2. Sodium sulphate is soluble in water but barium sulphate is sparingly soluble because :
(A) The hydration energy of Na_2SO_4 is more than its lattice energy.
(B) The lattice energy of $BaSO_4$ is more than its hydration.
(C) The lattice energy has no role to play in solubility.
(D) The lattice energy of Na_2SO_4 is more than its hydration energy.

Q3. Be and Al have following resemblance due to diagonal relationship:
(A) Have nearly equal electronegativity
(B) Form amphoteric oxides
(C) Have same charge/radius ratio
(D) Both form dimeric halides

Q4. Which of the following is are the characteristic of barium?
(A) It produce water soluble sulphide, sulphite and sulphate.
(B) It is a silvery white metal.
(C) It forms $Ba(NO_3)_2$ which is used in preparation of green fire.
(D) When CO_2 gas is passed into baryta water, it turns to milky.

Q5. Which of the following statements are false?
(A) $BeCl_2$ is a linear molecule in the vapour state but it is polymeric form in the solid state
(B) Calcium hydride is called hydrolith.
(C) Carbides of both Be and Ca react with water to form acetylene
(D) Oxides of both Be and Ca are amphoteric.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 6 to 8

On the basis of E.N. difference of $|X_0 - X_E|$ and $|X_0 - X_H|$, acidic and basic characters of E–O–H can be decided, and acidic and basic strengths are compared.

Q6. Correct order of acidic strength is :-
(A) Na–O–H > K–O–H > Rb–O–H
(B) Na–O–H < K–O–H < Li–O–H
(C) H–O–K > H–O–Rb > Na–O–H
(D) None of these

Q7. Correct order of basic strength is :-
(A) H–O–I < H–O–Br < H–O–Cl
(B) H–O–Cl > H–O–I > H–O–Br
(C) H–O–I > H–O–Br > H–O–Cl
(D) None of these

Q8. Strongest acid among the following :-
(A) HClO
(B) $HClO_2$
(C) $HClO_4$
(D) $HClO_3$

Paragraph for Q. 9 to 11

$A \xrightarrow{\Delta} B\ (oxide) + CO_2$

$B + H_2O \longrightarrow C$

$C + CO_2 \longrightarrow A\ (milky)$

$C + NH_4Cl \xrightarrow{\Delta} D\ (gas)$

$D + H_2O + CO_2 \longrightarrow E$

$E + NaCl \longrightarrow F$

$F \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

Q9. A is :

(A) $Ca(HCO_3)_2$ (B) $CaCO_3$

(C) CaO (D) Na_2CO_3

Q10. B and C are :

(A) CaO, $Ca(OH)_2$

(B) $Ca(OH)_2$, $CaCO_3$

(C) $CaCO_3$, $Ca(OH)_2$

(D) $Ca(OH)_2$, CaO

Q11. D, E and F are :

(A) NH_3, NH_4Cl, NH_4HCO_3

(B) NH_3, NH_4HCO_3, $NaHCO_3$

(C) NH_4HCO_3, Na_2CO_3, $NaHCO_3$

(D) None

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q12. Match list I with list II and choose the correct answer from the codes given below :

List I	**List II**
(A) $NaNO_3$	(a) Baking soda
(B) $Na(NH_4)HPO_4$	(b) Chile salt peter
(C) $NaHCO_3$	(c) Microcosmic salt
(D) $Na_2CO_3.10H_2O$	(d) Washing soda

Codes is :

	A	B	C	D
(1)	a	b	c	d
(2)	b	c	a	d
(3)	c	a	b	d
(4)	d	a	b	c

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13. Match the elements given in Column-I with the colour they impart to the flame given in Column-II.

Column-I	**Column-II**
(A) Cs	(P) Apple green
(B) Na	(Q) Violet
(C) K	(R) Brick red
(D) Ca	(S) Yellow
(E) Sr	(T) Crimson red
(F) Ba	(U) Blue

Q14. Match the column:

Column-I	**Column-II**
(A) Li	(P) Radioactive
(B) Na	(Q) Atomic number 11
(C) K	(R) Cation of element which has higher degree of hydration among the elements
(D) Fr	(S) ns^1 configuration
	(T) First elements of group-I

Q15. Match the compounds given in Column-I with their uses mentioned in Column-II.

Column-I	**Column-II**
(A) $CaCO_3$	(P) Dentistry, ornamental work
(B) $Ca(OH)_2$	(Q) Manufacture of sodium carbonate from caustic soda
(C) CaO	(R) Manufacture of high quality paper
(D) $CaSO_4$	(S) Used in white washing

Q16.

Column-I	**Column-II**
(A) Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(P) Ba
(B) Metal cation + K_2CrO_4 $\longrightarrow$ yellow ppt	(Q) Sr
(C) Metal + NH_3 $\xrightarrow{\text{(liquid)}}$ blue solution	(R) Na
(D) MCl_2 + conc. H_2SO_4 $\longrightarrow$ white ppt.	(S) Mg

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. What is the most common oxidation state of alkali metals?

Q18. Find the number of chloride(s) which do not impart color to the flame from the following:

$BeCl_2$, $SrCl_2$, $MgCl_2$, $CaCl_2$, $BaCl_2$

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The metal which reacts with water at room temperature is -
(A) Copper (B) Iron
(C) Magnesium (D) Sodium

Q2. Brine solution on electrolysis will not give :-
(A) NaOH (B) Cl_2
(C) O_2 (D) H_2

Q3. Sodium oxide (Na_2O) is produced when :
(A) Na_2CO_3 is heated
(B) Na burns with excess of O_2
(C) Na burns in limited O_2
(D) Na reacts with moist air

Q4. When element M of I A group reacts with water, it gives
(A) MOH + O_2 (B) MOH + H_2O
(C) MOH + H_2 (D) MH + H_2

Q5. The paramagnetic species is :–
(A) KO_2 (B) SiO_2
(C) TiO_2 (D) BaO_2

Q6. Select the correct statement :-
(a) $CaSO_4.2H_2O \xrightarrow{393K} CaSO_4\frac{1}{2}H_2O$
(b) $NaOH(\text{hot \& con.}) + Br_2 \longrightarrow NaBr + NaBrO_4 + H_2O$
(c) $Mg + N_2 \longrightarrow Mg_3N_2 \xrightarrow{+H_2O} NH_3(\text{gas})$
(d) $Be + C \longrightarrow Be_2C \xrightarrow{+H_2O} CH_4$
Correct answer is :-
(A) Only a and d (B) b, c, d
(C) a, b, c, d (D) a, c, d

Q7. The alkali metals which form normal oxide, peroxide as well as super oxides are
(A) Na, Li (B) K, Li
(C) Li, Cs (D) K, Rb

Q8. The aqueous solutions of lithium salts are poor conductor of electricity rather than other alkali metals because of
(A) high ionisation energy
(B) high electronegativity
(C) lower ability of Li^+ ions to polarize water molecules
(D) higher degree of hydration of Li^+ ions

Q9. An element A has configuration $1s^2 2s^2 2p^6 3s^2$ formula of its phosphate is :-
(A) ASO_4 (B) APO_4
(C) $A_3(PO_4)_2$ (D) $A_2(PO_4)_3$

Q10. Na_2CO_3 can be manufactured by Solvay's process but K_2CO_3 cannot be prepared because:-
(A) K_2CO_3 is more soluble
(B) K_2CO_3 is less soluble
(C) $KHCO_3$ is more soluble than $NaHCO_3$
(D) $KHCO_3$ is less soluble than $NaHCO_3$

Q11. Chile saltpetre is the ore of :-
(A) Iodine (B) Bromine
(C) Sodium (D) Magnesium

Q12. Which of the following is not alum :-
(A) $(NH_4)_2SO_4.Al_2(SO_4)_3.24H_2O$
(B) $(NH_4)_2SO_4.Fe_2(SO_4)_3.24H_2O$
(C) $(NH_4)_2SO_4.Zn_2(SO_4)_3.24H_2O$
(D) $K_2SO_4.Al_2(SO_4)_3.24H_2O$

Q13. Washing soda is
(A) $Na_2CO_3.7H_2O$
(B) $Na_2CO_3.10H_2O$
(C) $Na_2CO_3.H_2O$
(D) Na_2CO_3

Q14. $NaOH(Solid) + CO \xrightarrow{200°C} X$; product X is -

(A) $NaHCO_3$ (B) Na_2CO_3

(C) HCOONa (D) H_2CO_3

Q15. A colourless solid (X) on heating evolved CO_2 and also gave a white residue, soluble in water. Residue also give CO_2 when treated with dilute acid (X) is

(A) Na_2CO_3 (B) $CaCO_3$

(C) $NaHCO_3$ (D) $Ca(HCO_3)_2$

Q16. In diborane –

(A) 2 bridged hydrogen and four terminal hydrogen are present

(B) 3 bridged and three terminal hydrogen are present

(C) 4 bridged hydrogen atoms are not in the same plane in dibroane

(D) 1 bridged hydrogen and 1 terminal hydrogen are present

Q17. Which is most stable out of the following ?

(A) $[Be(H_2O)_4]^{2+}$ (B) $[Mg(H_2O)_4]^{2+}$

(C) $[Ca(H_2O)_4]^{2+}$ (D) $[Sr(H_2O)_4]^{2+}$

Q18. Order of solubility of carbonates of alkaline earth metal is :-

(A) $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3$

(B) $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4$

(C) $SrCO_3 > CaCO_3 > MgCO_3 > BeCO_3$

(D) $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3$

Q19. The lattice energy of the lithium is in the following order :

(A) LiF > LiCl > LiBr > LiI

(B) LiCl > LiF > LiBr > LiI

(C) LiBr > LiCl > LiF > LiI

(D) LiI > LiBr > LiCl > LiF

Q20. The hydration energy of Mg^{2+} is -

(A) more than that of Mg^{3+} ion

(B) more than that of Na^+ ion

(C) more than that of Al^{3+} ion

(D) more than that of Be^{2+} ion

Q21. $MgBr_2$ and MgI_2 are soluble in acetone because of

(A) Their ionic nature

(B) Their coordinate nature

(C) Their metallic nature

(D) Their covalent nature

Q22. The complex formation tendency of alkaline earth metals decreases down the group because

(A) atomic size increases

(B) availability of empty d and f-orbitals increases

(C) nuclear charge to volume ratio increases

(D) all the above

Q23. Gypsum on heating, not gives :-

(A) $CaSO_4 . \frac{1}{2}H_2O$ (B) $CaSO_4$

(C) $CaO + SO_3$ (D) $CaS + O_2$

Q24. When Cl_2 is passed into moist slaked lime, compound formed is :

(A) CaO_2Cl_2 (B) CaO_2Cl

(C) $CaOCl_2$ (D) $CaCl_2O_4$

Q25. Gypsum $CaSO_4.2H_2O$ on heating to about 120°C forms plaster of paris, which has chemical composition represented by :-

(A) $2CaSO_4. 3H_2O$ (B) $CaSO_4.H_2O$

(C) $CaSO_4. \frac{1}{2}H_2O$ (D) $CaSO_4. 1\frac{1}{2}H_2O$

Q26. On heating anhydrous Na_2CO_3.....is evolved:-

(A) CO_2 (B) Water vapour

(C) CO (D) No gas

Q27. The formula of sodium zeolite which is used in permutit process for softening water is

(A) $Na_2O.Al_2O_3.Si_2O_4.xH_2O$

(B) $Na_2.Al_2.Si_2O_4.xH_2O$

(C) $Na_2O.AlO_3.SiO_4.xH_2O$

(D) $K_2Al_2SiO_8.xH_2O$.

Q28. Which of the following statement is **incorrect**:

(A) K_2CO_3 can also be prepared by Solvay process

(B) Li_2CO_3 does not exist in solid form

(C) Be is inert towards reaction with water

(D) Gypsum slow down the process of setting of cement

Q29. $Na + Al_2O_3 \xrightarrow{\text{High temperature}} X \xrightarrow[\text{water}]{CO_2 \text{ in}} Y$; compound Y is -

(A) $NaAlO_2$ (B) $NaHCO_3$

(C) Na_2CO_3 (D) Na_2O_2

Q30. (Milky Cloud) $C \xleftarrow{CO_2} A + Na_2CO_3 \longrightarrow B + C$

The chemical formulae of A and B are

(A) NaOH and $Ca(OH)_2$

(B) $Ca(OH)_2$ and NaOH

(C) NaOH and CaO

(D) CaO and $Ca(OH)_2$

Q31. $X \xrightarrow[\text{Catalyst}]{CoCl_2} CaCl_2 + Y \uparrow$; the effective ingredient of X is -

(A) OCl^- (B) Cl^-

(C) OCl^+ (D) OCl_2^-

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Q1. When dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does occur?

(A) Blue colored solution is obtained.

(B) Na^+ ions are formed in the solution.

(C) Liquid NH_3 becomes good conductor of electricity.

(D) Liquid NH_3 remains diamagnetic.

Q2. Which of the following element(s) is/are not react directly with nitrogen to form nitride?

(A) Na (B) Li

(C) K (D) Rb

Q3. KO_2 finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO_2 is/are :

(A) it produces O_2

(B) it produces O_3

(C) it absorbs CO_2

(D) it absorbs both CO and CO_2

Q4. The compound(s) which have –O–O– bond(s) is/are

(A) BaO_2 (B) Na_2O_2

(C) CrO_5 (D) Fe_2O_3

Q5. Select the incorrect order for given properties :

(A) Thermal stability :
$BaSO_4 > SrSO_4 > CaSO_4$

(B) Solubility :
$BaSO_4 > SrSO_4 > CaSO_4$

(C) Thermal stability :
$Li_2CO_3 < Na_2CO_3 < K_2CO_3$

(D) Solubility :
$Li_2CO_3 > Na_2CO_3 > K_2CO_3$

Q6. Identify the incorrect statement :

(A) Gypsum contains a lower percentage of calcium than Plaster of Paris

(B) Gypsum is obtained by heating Plaster of Paris

(C) Plaster of Paris is obtained by hydration of gypsum

(D) Plaster of Paris is obtained by partial oxidation of gypsum

Q7. Which one of the following processes is/are not used for the manufacture of calcium?

(A) Reduction of CaO with carbon

(B) Reduction of CaO with hydrogen

(C) Electrolysis of a mixture of anhydrous $CaCl_2$

(D) Electrolysis of molten $Ca(OH)_2$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.8 to 12

Alkali and alkaline earth metals along with hydrogen and helium constitute s-block elements. They have low ionization enthalpies and hence exhibit characteristic flame colorations. They have highly negative electrode potentials and hence are strong reducing agents. Their solutions in liquid ammonia are conducting and also act as strong reducing agents. Being stronger reducing agents than hydrogen, they are usually prepared by electrolysis of their fused chlorides. Their oxides are basic and the basic strength increases down the group. The solubility of carbonates and sulphates of alkali and alkaline earth metals shows opposite trends. The carbonates of alkaline earth metals and lithium carbonate decompose on heating while the carbonates of other alkali metals do not decompose on heating. The bicarbonates of both alkali and alkaline earth metals on heating give carbonates.

Q8. The compound insoluble in acetic acid is:
(A) Calcium oxide
(B) Calcium carbonate
(C) Calcium oxalate
(D) Calcium hydroxide

Q9. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order: K_2CO_3(I), $MgCO_3$(II), $CaCO_3$(III), $BeCO_3$(IV) :-
(A) I < II < III < IV
(B) IV < II < III < I
(C) IV < II < I < III
(D) II < IV < III < I

Q10. Property of alkaline earth metals that increases with their atomic number is:
(A) Ionization energy
(B) Solubility of their hydroxides
(C) Solubility of their sulphates
(D) Electronegativity

Q11. Which of the following process is used in the extractive metallurgy of magnesium?
(A) Fused salt electrolysis
(B) Self-reduction
(C) Aqueous solution electrolysis
(D) Thermite reduction

Q12. Identify the correct order of acidic strengths of CO_2, CuO, CaO, H_2O
(A) CaO < CuO < H_2O < CO_2
(B) H_2O < CuO < CaO < CO_2
(C) CaO < H_2O < CuO < CO_2
(D) H_2O < CO_2 < CaO < CuO

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13.

Column-I	**Column-II**
(P) Solvay process used for	(A) NaCl
(Q) Evolve CO_2 ↑ on heating	(B) Na_2O_2
(R) aq. soln. is neutral towards litmus	(C) $NaHCO_3$
(S) Ozone oxidises	(D) Na_2CO_3

Q14. Match the column:

Column-I	**Column-II**
(A) $Li^+(aq)$	(P) Metal(s) readily react vigorously with halogens to from ionic halides
(B) $Na^+(aq)$	(Q) Highest ionic mobility among the ions
(C) $K^+(aq)$	(R) Highest hydrated size among the ions
(D) $Rb^+(aq)$	(S) Highest hydration energy among the cation
	(T) Lowest ionic mobility among the ions

Q15. Match the column:

Column-I	**Column-II**
(A) Alkali metal carbonate which does not decompose on heating	(P) $MgCO_3$
(B) Alkali metal carbonate which decomposes on heating	(Q) Na_2CO_3
(C) K/liq. NH_3 is blue color solution which conducts electricity	(R) Li_2CO_3
	(S) Solvated electrons

TRIPLE MATCH

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I	**Column-II**	**Column-III (Colour in Flame test)**
(a) Be	(p) Show diagonal relationship with other element	(x) Apple green
(b) Mg	(q) Highest hydration energy of cation among element	(y) Brick red
(c) La	(r) Cation has lowest hydrated size in aquous solution	(z) Crimson
(d) Sr	(s) Oxide and hydroxide are amphoteric	(w) Doesn't impart any colour to the flame

Q16. Which of the following options is the correct combination?

(A) (a)(q)(w) (B) (a)(s)(r) (C) (b)(p)(x) (D) (d)(r)(w)

Q17. Correct combination for the element whose maximum co-ordination no. is 4, is :-

(A) (b)(p)(w) (B) (d)(r)(z) (C) (a)(s)(w) (D) (a)(p)(r)

Q18. Correct combination for element which shows diagonal relationship with Li is :-

(A) (c)(p)(w) (B) (b)(p)(w) (C) (a)(q)(r) (D) (b)(s)(w)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. Find the number of element(s) which is/are present in group-I :

Be, Li, Mg, Al, Na, Ca, Cs

Q20. Find the number of compounds from the following in which the element in the anionic part is in the minimum oxidation state of it LiH, Mg_3Bi_2, Al_4C_3, Ca_3P_2, BaO_2.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	B	D	C	D	C	B	A	A	C
11	12	13	14	15	16	17	18	19	20
C	A	A	A	A	A	D	C	D	D
21	22	23	24	25	26	27	28	29	30
B	C	A	A	A	B	A	D	B	C
31	32	33	34	35	36	37	38	39	40
D	A	C	B	C	C	B	D	A	A
41	42	43	44	45	46	47	48	49	50
B	A	D	B	D	B	D	C	C	B

Ex-1A

1. A, B, D **2.** A, B **3.** A, B, C, D **4.** B, C, D **5.** C, D
6. A **7.** C **8.** C **9.** B **10.** A
11. B **12.** B
13. (A) → U; (B) → S; (C) → Q; (D) → R; (E) → T; (F) → P
14. (A) → R,S,T; (B) → Q,S; (C) → S; (D) → P,S
15. (A) → R; (B) → S; (C) → Q; (D) → P
16. (A) → P,Q,S ; (B) → P,Q; (C) → P,Q,R ; (D) → P,Q
17. 1 **18.** 2

Ex-2									
1	2	3	4	5	6	7	8	9	10
D	C	C	C	A	D	D	D	C	C
11	12	13	14	15	16	17	18	19	20
C	C	B	C	C	A	A	D	A	B
21	22	23	24	25	26	27	28	29	30
D	A	D	C	C	D	A	A	C	B
31									
A									

Ex-2A

1. A, B, C **2.** A, C, D **3.** A, C **4.** A, B, C **5.** B, D
6. B, C, D **7.** A, B, D **8.** C **9.** B **10.** B
11. A **12.** A **13.** (P) → (D); (Q) → (C); (R) → (A); (S) → (D)
14. (A) → R,S,T; (B) → P; (C) → P; (D) → P,Q
15. (A) → Q; (B) → R; (C) → S
16. 2 **17.** 4 **18.** A **19.** C **20.** B

-: Important Notes & Formulas :-